Rising Skill Premium and the Innovation on Tools

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Progress

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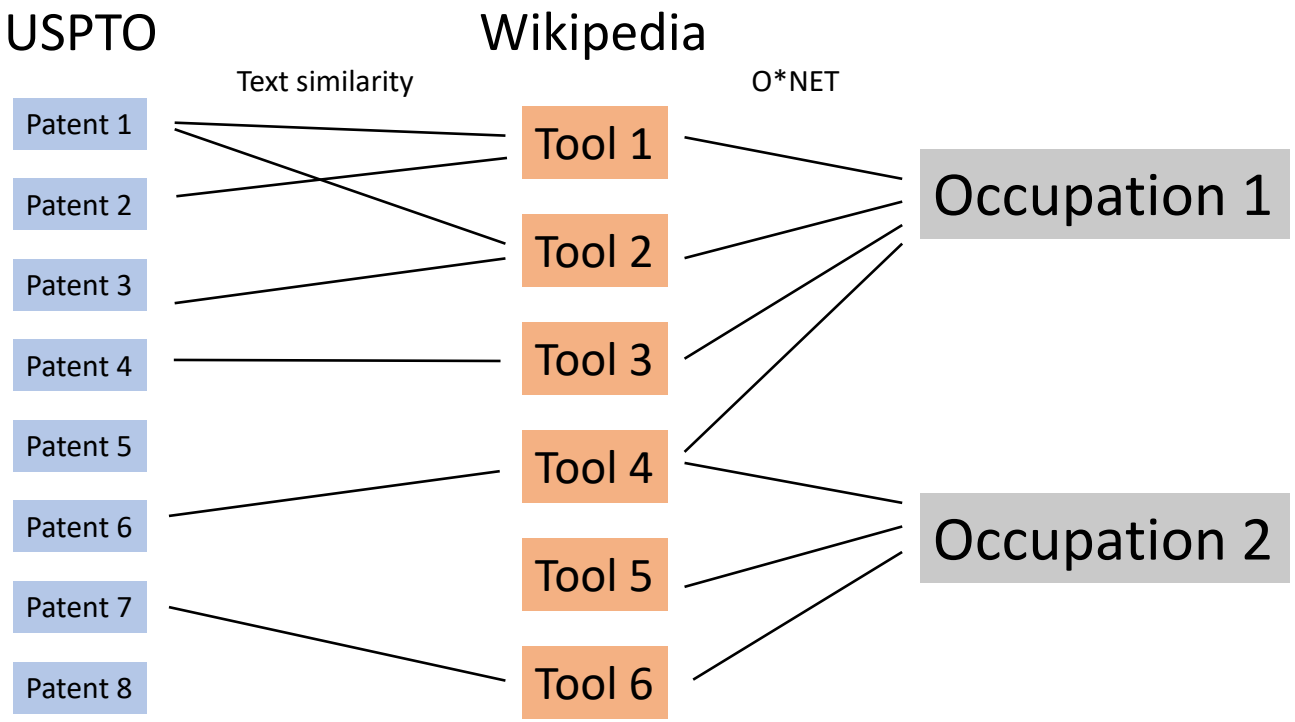
Motivation and Research Question

- Rising skill premium since 1980 graph
- Main hypothesis in the literature: Skill-biased technical change
  - Development of equipment↑ + complement with skilled workers → productivity of skilled worker ↑ → skill premium ↑
- **This paper:** Using text analysis, measure innovation on tools
  - Construct tool innovation measure at occupation level
  - It contributed to rising skill premium by
    - Increase wage of skill-intensive jobs
    - Increase emp share of high skill premium jobs
    - Increase skill premium/intensity within occupation
- **Final goal:** Quantify the contribution of tool innovation on increasing skill premium in a general equilibrium Compare with Webb's JMP

Summary of datawork



Simple Model

- Competitive labor market by occupation and skill level.
- Occupation-specific tasks are needed to produce aggregate output.

$$Y_t = Z_t \left(\sum_o \alpha_{ot} \left(\lambda_{ot} \left(x_{ot} s_{ot} \right)^{\frac{\sigma-1}{\sigma}} + (1-\lambda_{ot}) \left({x_{ot}}^{\delta} u_{ot} \right)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}}$$

  - α_{ot} - task demand shifter
  - x_{ot} - productivity shifter
  - δ - Loading on the unskilled labor relative to skilled labor
  - λ_{ot} - occupation-specific skilled labor demand shifter
  - σ, ρ - within and across elasticity of substitution
- **Objective:** Directly measure x_{ot} from patent
- Get the contribution of x_{ot} in rising skill premium

Data

Data: Tools

- Tool - occupation data from ONET
  - 775 occupations, each have 39 tools on average.
  - 4,180 unique tools in total

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Occupation	Tools
Aerospace Engineers	Lasers
Aerospace Engineers	Atomic force microscopes
Aerospace Engineers	Aircraft engine compressors
Aerospace Engineers	Turbine engines Vibration testers
Aerospace Engineers	Power grinders
Aerospace Engineers	Pressure sensors
Bakers	Automatic dough presses
Bakers	Bagel dividers
Bakers	Baking frames
Bakers	Baking sheets
Bakers	Balance scales
Bakers	Barquette molds

Data: Innovation

- United States Patent and Trademark Office (USPTO)
  - Universe data of patent registered in the U.S.
  - Title, abstract, technology field
  - 6.1M utility patents in 1970-2015

Data: Wage, College Premium, Employment

- Decennial Census 1990, American Community Survey 2015-2019
- Employment and average wage by occupation, year, industry, and education group (College+ vs. Below)

⇒ College premium and skill intensity at the occupation level.

Table: Summary Statistics of changes in 1990-2015

	Mean	Sd	Median	1Q.	3Q.	N.
Employment Growth	0.24	1.07	0.08	-0.26	0.60	420
Wage Growth	1.11	0.47	1.01	0.84	1.23	420
D. College Premium (%)	0.48	37.32	3.96	-7.68	16.69	405
D. Skill Intensity (%)	3.63	5.64	3.05	0.38	5.88	420

Text Matching Procedure

Patent - Tool Matching

- **Objective**: Find related tools to each patent
- Patent: Title, abstract
- Tools: Use wikipedia page of tools
- Calculate text similarity and match each patent to tools
  - Lemmatize text data and drop stopwords (e.g., is, the)
  - Vectorize text and weight by the inverse of frequency of words
  - Calculate cosine similarity of bigrams between two texts
  - Apply 0.025 as threshold to be matched (changed for robustness check)
  - For each patent, allow at most 5 tools matched and weighted by the inverse number of tools matched

Example of Matching

Patent: System and method for detecting deterioration of oxygen sensor

feedback type air-fuel ratio control system control air-fuel ratio air-fuel mixture fed internal combustion engine accordance information signal issued first oxygen sensor installed exhaust line engine exhaust line catalytic converter position downstream first oxygen sensor provided system control system detects deterioration first oxygen sensor

Wikipedia: Oxygen sensor

oxygen sensor lambda sensor lambda refers air-fuel equivalence ratio usually denoted electronic device measure proportion oxygen gas liquid analysed common application measure exhaust gas concentration oxygen internal combustion engine automobile vehicle order calculate required dynamically adjust air-fuel ratio catalytic converter work optimally.

Similarity: 0.25. Oxygen sensor is used by biochemical engineers

Summary Statistics of Matching Result

- Wikipedia page is found for 1,825 / 4,180 tools
- 4.5M patents are matched with tools; 25% cannot be matched
- We calculate average number of patents at occupation level example

Table: The number of patent matched to each tool

	N	mean	median	SD
Patent (70-90)	1,824	456	238	610
Patent (90-15)	1,824	2,048	904	3,233

Table: The number of patent matched to each occupation

	N	mean	median	SD
Patent (70-90)	775	361	346	172
Patent (90-15)	775	2,267	1,926	1,309

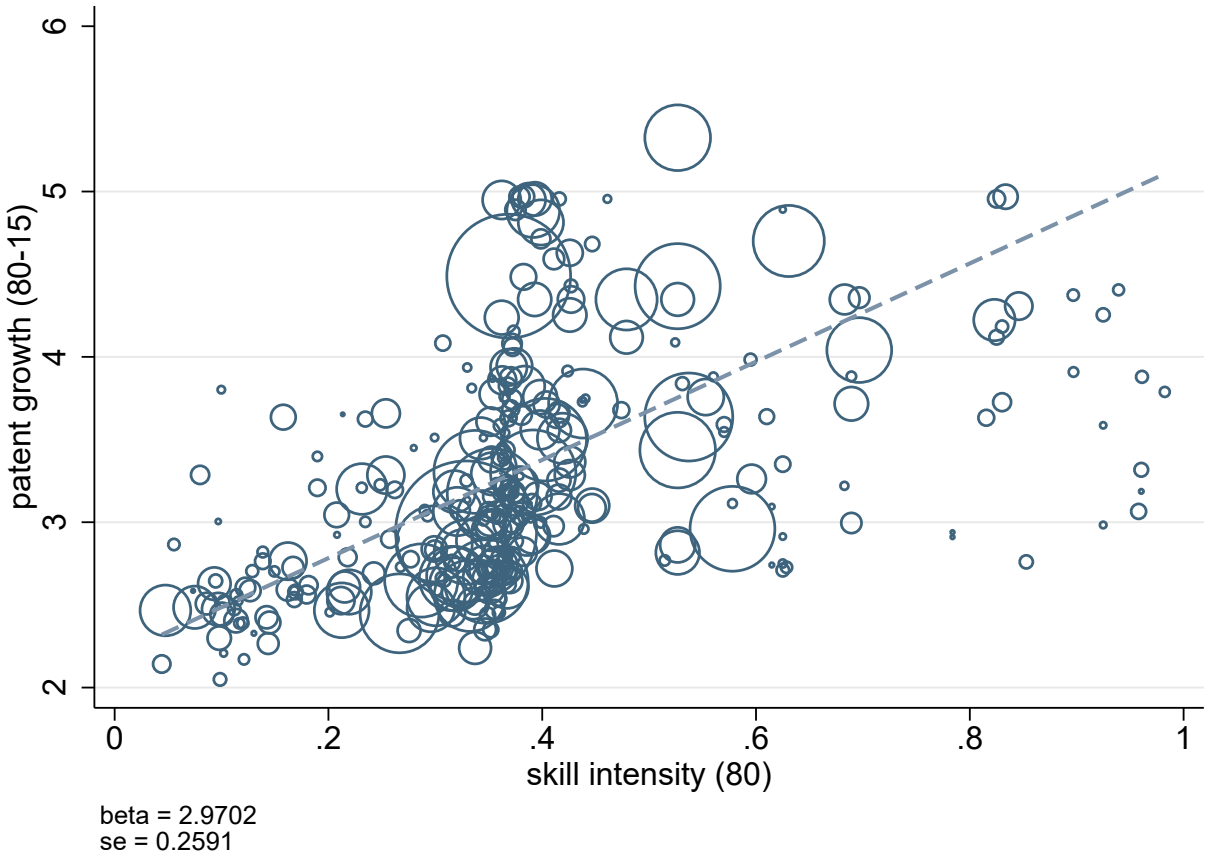
histogram

Top and bottom occupations with patent growth

- Top 5 occupation : Legislators, Film and Video Editors, Web Developers, First-Line Supervisors of Non-Retail Sales Workers, Desktop Publishers
- Bottom 5 occupation : Tailors, Textile Knitting and Weaving Machine Setters, Fabric Menders, Sewing Machine Operators

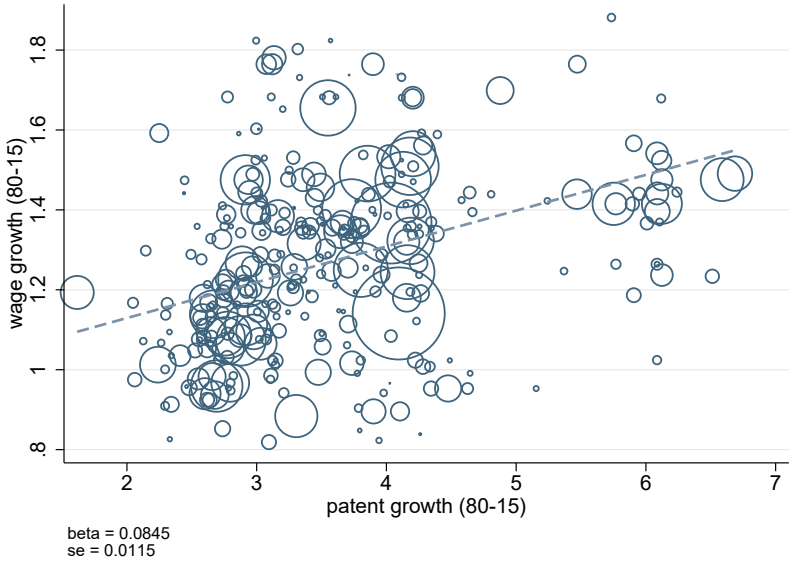
Reduced Form Analysis

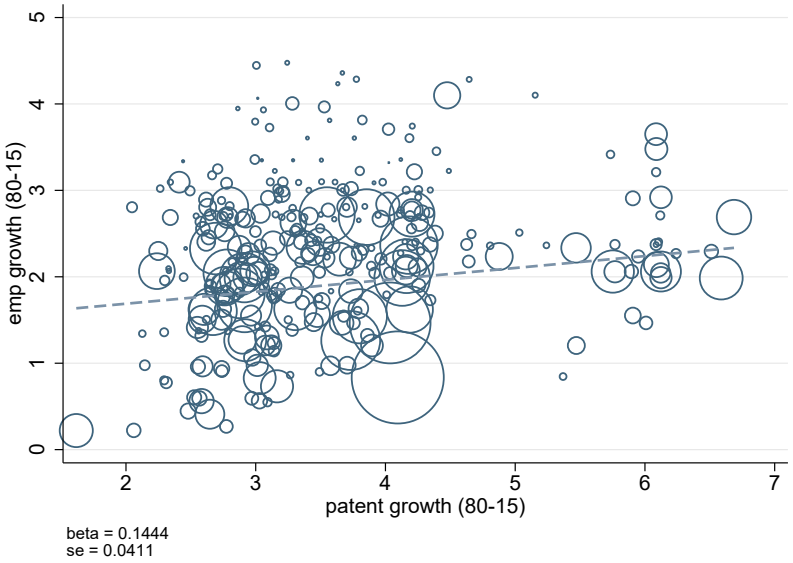
Patent Grows More in Skill-intensive Jobs



Skill intensity = skilled labor share at occupation level. Skilled = College graduate or higher. The size of circle is proportional to wage bill at the initial year.

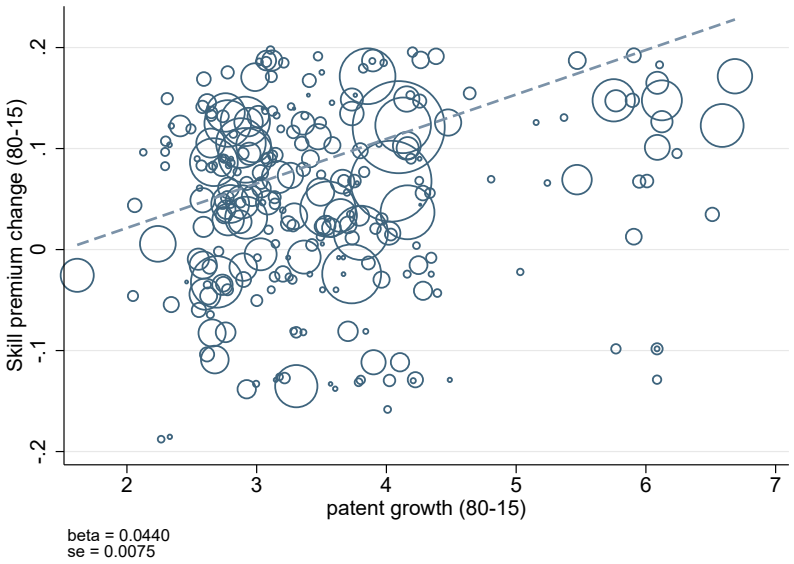
Tool innovation increases wage and employment

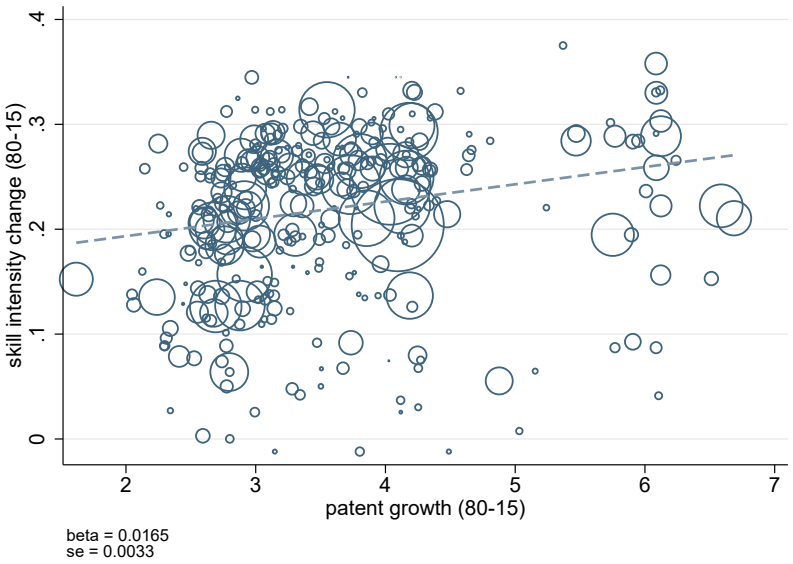




The size of circle is proportional to wage bill at the initial year.

Tool Innovation increases relative wage and employment of skilled workers





Skill premium = Average wage of skilled / Average wage of unskilled at occupation level.
Skilled = College graduate or higher. weighted by wage bill at the initial year.

Regression Result

$$\Delta y_{i,1990-2015} = \beta_0 + \beta_1 \Delta \log p_{i,1990-2015} + \epsilon_i$$

	(1) Wage	(2) Emp.	(3) Skill Pre.	(4) Skill Int.
patent	0.0845^{***}	0.144^{***}	0.0440^{***}	0.0165^{***}
	(0.0115)	(0.0411)	(0.00753)	(0.00329)
N	324	324	323	324

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Wage and employment is logged. Wage and college premium regression is weighted by wage bill in the initial period. with market value other specifications

Instrumental Variables for Innovation on Tools

- Instrumental variable for x_{ot}
- Concern : sectors with positive demand shock would have larger increase in innovation - reverse causality problem
  - e.g., innovation in computer sector $\rightleftarrows$ wage increase in IT sector
- Use technology sector that has knowledge spillover to the interested sector
  - Innovation in electricity is correlated with innovation in computer sector but not with demand shock of IT workers
  - Use the number of patents in “upstream" sector as an instrument
  - **Assumption:** When patent A cites B, knowledge spillover from B to A
  - Use patent citation data, construct the share of citations in patent class level, identify “upstream" sector
  - We exclude technology sectors that are in the same NAICS 2 digit

Construction of IV

$$\sum_{j \neq i} s_{ij} \Delta \log P_{jt} \quad \rightarrow \quad \Delta \log P_{it}$$

- s_{ij} is the citation share from technology class i to j

IV Regression Result

$$\Delta y_{i,1990-2015} = \beta_0 + \beta_1 \Delta p_{i,1990-2015} + \epsilon_i$$

	(1) Wage	(2) Emp.	(3) Skill Pre.	(4) Skill Int.
patent	0.119^{***}	0.0600	0.0406^{***}	0.0297^{***}
	(0.0156)	(0.0551)	(0.0100)	(0.00449)
N	324	324	323	324
F (first stage)	410.8	410.8	409.6	410.8

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Wage and employment is logged. Wage and college premium is weighted by wage bill in the initial period.

first stage with market value other specifications

★ How? Innovation reduces the price for tools. Innovation and Tool Price

Structural Estimation

Summary of Structural Estimation

- Build a model with labor demand + supply
- Use tool innovation measure as productivity at occupation level, potentially different impact on skilled / unskilled workers
- Get other variables such as unknown demand shock for skilled worker within occupation and across occupation.
- Counterfactual analysis: Change only tool innovation variable - how much can we explain the increased skill premium?
- How much comes from within-occupation margin? or across-occupation margin?

Representative Firm's Problem

- Competitive labor market by occupation and skill level.
- Occupation-specific tasks are needed to produce aggregate output.

$$Y_t = Z_t \left(\sum_o \alpha_{ot} \left(\lambda_{ot} \left(x_{ot} s_{ot} \right)^{\frac{\sigma-1}{\sigma}} + (1-\lambda_{ot}) \left(x_{ot}^{\delta} u_{ot} \right)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}}$$

- subject to

$$C_t = \sum w_{ot}^u u_{ot} + \sum w_{ot}^s s_{ot}$$

  - α_{ot} - task demand shifter of occupation o at time t
  - x_{ot} - productivity shifter
  - δ - Loading on the unskilled labor relative to skilled labor
  - λ_{ot} - occupation-specific skilled labor demand shifter
  - σ, ρ - within- and across- elasticity of substitution

First Order Conditions

- Rearranging the first order conditions gives

$$\frac{w_{ot}^s}{w_{ot}^u} = \left(\frac{\lambda_{ot}}{1-\lambda_{ot}}\right)\left(x_{ot}^{1-\delta}\right)^{\frac{\sigma-1}{\sigma}}\left(\frac{s_{ot}}{u_{ot}}\right)^{-\frac{1}{\sigma}}$$

$$\frac{w_{ot}^s}{w_{pt}^s} = \frac{\lambda_{ot}}{\lambda_{pt}}\left(\frac{x_{ot}}{x_{pt}}\right)^{\frac{\sigma-1}{\sigma}}\frac{\alpha_{ot}}{\alpha_{pt}}\left(\frac{\theta_{ot}}{\theta_{pt}}\right)^{\frac{\rho-\sigma}{\rho(\sigma-1)}}\left(\frac{s_{ot}}{s_{pt}}\right)^{\frac{\rho-\sigma}{\rho\sigma}}$$

$$\theta_{ot} := \left(\lambda_{ot} x_{ot}^{\frac{\sigma-1}{\sigma}} + (1-\lambda_{ot})\, x_{ot}^{\frac{(\sigma-1)(\delta\sigma-\sigma+1)}{\sigma}}\left(\frac{1-\lambda_{ot}}{\lambda_{ot}}\frac{w_{ot}^s}{w_{ot}^u}\right)^{\sigma-1}\right)$$

Occupation-Specific Labor Supply

- The occupation-specific labor supply function is

$$s_{ot} = \left(\frac{\exp(\log w^s_{ot} + \epsilon_{ot})}{\sum_{o'} \exp(\log w^s_{o't} + \epsilon^s_{o't})} \right)^{\eta} L^s_t$$

$$u_{ot} = \left(\frac{\exp(\log w^u_{ot} + \epsilon_{ot})}{\sum_{o'} \exp(\log w^u_{o't} + \epsilon^u_{o't})} \right)^{\eta} L^u_t$$

GMM Estimation Equations

- Take log-differences on the demand and supply equations.

$$\Delta_t \log\left(\frac{w^s_{ot}}{w^u_{ot}}\right) = \Delta_t \log\left(\frac{\lambda_{ot}}{1-\lambda_{ot}}\right) + \frac{\sigma-1}{\sigma}(1-\delta)\,\Delta_t \log(x_{ot}) - \frac{1}{\sigma}\Delta_t \log\left(\frac{s_{ot}}{u_{ot}}\right)$$

$$\Delta_t \log\left(\frac{w^s_{ot}}{w^s_{1t}}\right) = \Delta_t \log\left(\frac{\lambda_{ot}}{\lambda_{pt}}\frac{\alpha_{ot}}{\alpha_{1t}}\right) + \frac{\rho-\sigma}{\rho(\sigma-1)}\Delta_t \log\left(\frac{\theta_{ot}}{\theta_{1t}}\right)$$
$$+ \frac{\sigma-1}{\sigma}\Delta_t \log\left(\frac{x_{ot}}{x_{1t}}\right) + \frac{\rho-\sigma}{\rho\sigma}\Delta_t \log\left(\frac{s_{ot}}{s_{1t}}\right)$$

$$\Delta_t \log\left(\frac{s_{ot}}{s_{1t}}\right) = \eta\left(\Delta_t \log\left(\frac{w^s_{ot}}{w^s_{1t}}\right) + \Delta_t\epsilon^s_{ot} - \Delta_t\epsilon^s_{1t}\right)$$

$$\Delta_t \log z^1_{ot}, \Delta_t \log z^2_{ot} \perp \Delta_t \log\left(\frac{\lambda_{ot}}{1-\lambda_{ot}}\right), \Delta_t \log\left(\frac{\lambda_{ot}}{\lambda_{pt}}\frac{\alpha_{ot}}{\alpha_{1t}}\right)$$

$$\Delta_t \log x_{ot} \perp (\Delta_t\epsilon^s_{ot}, \Delta_t\epsilon^u_{ot})$$

- We can estimate σ, δ, ρ, and η using GMM. detail
- We can back out α_{ot}, λ_{ot}, ϵ^s_{ot}, and ϵ^u_{ot}.

Instrumental Variables for GMM Estimation

- Instrumental variable for x_{ot}
- Concern : sectors with positive demand shock would have larger increase in innovation - reverse causality problem
  - e.g., innovation in computer sector $\rightleftarrows$ wage increase in IT sector
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  - **Assumption:** When patent A cites B, knowledge spillover from B to A
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Instrumental Variables for GMM Estimation

- Instrumental variable for s_{ot}/u_{ot}.
- Concern: demand shock for skilled worker could increase supply of skilled worker
- Use college graduates in different major to construct a Bartik-style supply shifter.
  - Using ACS 2009-2013, calculate the composition of college majors for each occupation
  - Using National Survey of College Graduates, calculate the growth rates for college graduates at the major level between 1993-2013
  - The Bartik supply shifter is calculated by

$$\log z_{ot}^2 = \sum_m \omega_{m,o}^{2009} (\log n_m^{2013} - \log n_m^{1993})$$

GMM Estimation Equations

- Take log-differences on the demand and supply equations.

$$\Delta_t \log\left(\frac{w^s_{ot}}{w^u_{ot}}\right) = \Delta_t \log\left(\frac{\lambda_{ot}}{1-\lambda_{ot}}\right) + \frac{\sigma-1}{\sigma}(1-\delta)\,\Delta_t \log(x_{ot}) - \frac{1}{\sigma}\Delta_t \log\left(\frac{s_{ot}}{u_{ot}}\right)$$

$$\Delta_t \log\left(\frac{w^s_{ot}}{w^s_{1t}}\right) = \Delta_t \log\left(\frac{\lambda_{ot}}{\lambda_{pt}}\frac{\alpha_{ot}}{\alpha_{1t}}\right) + \frac{\rho-\sigma}{\rho(\sigma-1)}\Delta_t \log\left(\frac{\theta_{ot}}{\theta_{1t}}\right)$$
$$+ \frac{\sigma-1}{\sigma}\Delta_t \log\left(\frac{x_{ot}}{x_{1t}}\right) + \frac{\rho-\sigma}{\rho\sigma}\Delta_t \log\left(\frac{s_{ot}}{s_{1t}}\right)$$

$$\log\left(\frac{s_{ot}}{s_{1t}}\right) = \eta\left(\log\left(\frac{w^s_{ot}}{w^s_{1t}}\right) + \epsilon^s_{ot} - \epsilon^s_{1t}\right)$$

$$\Delta_t \log z^1_{ot}, \Delta_t \log z^2_{ot} \perp \Delta_t \log\left(\frac{\lambda_{ot}}{1-\lambda_{ot}}\right), \Delta_t \log\left(\frac{\lambda_{ot}}{\lambda_{pt}}\frac{\alpha_{ot}}{\alpha_{1t}}\right)$$

$$\log x_{ot} \perp (\epsilon^s_{ot}, \epsilon^u_{ot})$$

- Given estimates for σ and λ_{ot} from the first regression, estimate ρ.
  - The same upstream IV is used in estimating the labor demand equation across occupations.
- Estimating the labor supply equation uses $\Delta_t \log\left(\frac{x_{ot}}{x_{1t}}\right)$.

Results

- Estimates for δ, σ, ρ, and η.

Parameter	δ	σ	ρ	η
Estimate	0.94	11.11	1.40	2.40
SE	(0.03)	(10.47)	(0.13)	(0.64)

Table: GMM Estimates

Counterfactual Analysis

- Goal: quantify the contribution of changes in x_{ot} on the rise in skill premium.
- Solving for the relative labor supply and skill premium for each occupation,

$$\log\left(\frac{w^s_{ot}}{w^u_{ot}}\right) = \log\left(\frac{\lambda_{ot}}{1-\lambda_{ot}}\right) + \frac{\sigma-1}{\sigma}(1-\delta)\log(x_{ot}) - \frac{1}{\sigma}\log\left(\frac{s_{ot}}{u_{ot}}\right)$$

$$\log\left(\frac{s_{ot}}{u_{ot}}\right) = \underbrace{\kappa_{ot}}_{=log\left(L^s_t/L^u_t\right)+\eta\epsilon^s_{ot}-\eta\epsilon^u_{ot}} + \eta\log\left(\frac{w^s_{ot}}{w^u_{ot}}\right)$$

$$\Rightarrow \log\left(\frac{w^s_{ot}}{w^u_{ot}}\right) = \left(1+\frac{\eta}{\sigma}\right)^{-1}\left\{\log\left(\frac{\lambda_{ot}}{1-\lambda_{ot}}\right) + \frac{\sigma-1}{\sigma}(1-\delta)\log(x_{ot}) - \frac{1}{\sigma}\kappa_t\right\}$$

$$\Rightarrow \log\left(\frac{s_{ot}}{u_{ot}}\right) = \kappa_{ot} + \eta\log\left(\frac{w^s_{ot}}{w^u_{ot}}\right)$$

- Estimation gives α_{ot}, λ_{ot}, and κ_{ot} as residuals.
- Plug in values of x_{ot} after 1990 and keep α_{ot}, λ_{ot}, and κ_{ot} at their levels before 1990.

Counterfactual Analysis

- Calculate the counterfactual values of relative employment and wage of skilled labor using the system of equations.

$$\log\left(\frac{w_{ot}^s}{w_{1t}^s}\right) = \log\left(\frac{\lambda_{ot}}{\lambda_{pt}}\frac{\alpha_{ot}}{\alpha_{1t}}\right) + \frac{\rho-\sigma}{\rho(\sigma-1)}\log\left(\frac{\theta_{ot}}{\theta_{1t}}\right) + \frac{\sigma-1}{\sigma}\log\left(\frac{x_{ot}}{x_{1t}}\right) + \frac{\rho-\sigma}{\rho\sigma}\log\left(\frac{s_{ot}}{s_{1t}}\right)$$

$$\log\left(\frac{s_{ot}}{s_{1t}}\right) = \eta\left(\log\left(\frac{w_{ot}^s}{w_{1t}^s}\right) + \epsilon_{ot}^s - \epsilon_{1t}^s\right)$$

$$\Rightarrow \log\left(\frac{w_{ot}^s}{w_{1t}^s}\right) = \left(1 - \frac{(\rho-\sigma)\eta}{\rho\sigma}\right)^{-1}\left(\log\left(\frac{\lambda_{ot}}{\lambda_{pt}}\frac{\alpha_{ot}}{\alpha_{1t}}\right) + \frac{\rho-\sigma}{\rho(\sigma-1)}\log\left(\frac{\theta_{ot}}{\theta_{1t}}\right) + \frac{\sigma-1}{\sigma}\log\left(\frac{x_{ot}}{x_{1t}}\right) + \frac{(\rho-\sigma)\eta}{\rho\sigma}\left(\epsilon_{ot}^s - \epsilon_{1t}^s\right)\right)$$

$$\Rightarrow \log\left(\frac{s_{ot}}{s_{1t}}\right) = \eta\left(\log\left(\frac{w_{ot}^s}{w_{1t}^s}\right) + \epsilon_{ot}^s - \epsilon_{1t}^s\right)$$

- Using residuals $\log(\alpha_{ot}/\alpha_{1t})$ and $\epsilon_{ot}^s - \epsilon_{1t}^s$ of the pre-1990 period, calculate $\log\left(\frac{w_{ot}^s}{w_{1t}^s}\right)$ and $\log\left(\frac{s_{ot}}{s_{1t}}\right)$ after changes in only x_{ot}.

Counterfactual Analysis

- The college premium can be expressed as

$$
\begin{aligned}
college\,premium &= \frac{\frac{1}{L_t^s}\sum s_{ot} w_{ot}^s}{\frac{1}{L_t^u}\sum u_{ot} w_{ot}^u} \\
&= \frac{L_t^u}{L_t^s}\frac{s_{1t} w_{1t}^s \sum \frac{s_{ot}}{s_{1t}} \frac{w_{ot}^s}{w_{1t}^s}}{s_{1t} w_{1t}^s \sum \frac{u_{ot}}{s_{ot}} \frac{s_{ot}}{s_{1t}} \frac{w_{ot}^u}{w_{ot}^s} \frac{w_{ot}^s}{w_{1t}^s}}. \\
&= \frac{L_t^u}{L_t^s}\frac{\sum \frac{s_{ot}}{s_{1t}} \frac{w_{ot}^s}{w_{1t}^s}}{\sum \frac{u_{ot}}{s_{ot}} \frac{s_{ot}}{s_{1t}} \frac{w_{ot}^u}{w_{ot}^s} \frac{w_{ot}^s}{w_{1t}^s}}
\end{aligned}
$$

  which means we can normalize $s_{1t} w_{1t}^s$ to one.

- We fix L_t^u / L_t^s at the level of pre-1990 period as κ_{ot} encapsulates L_t^u / L_t^s.

Results

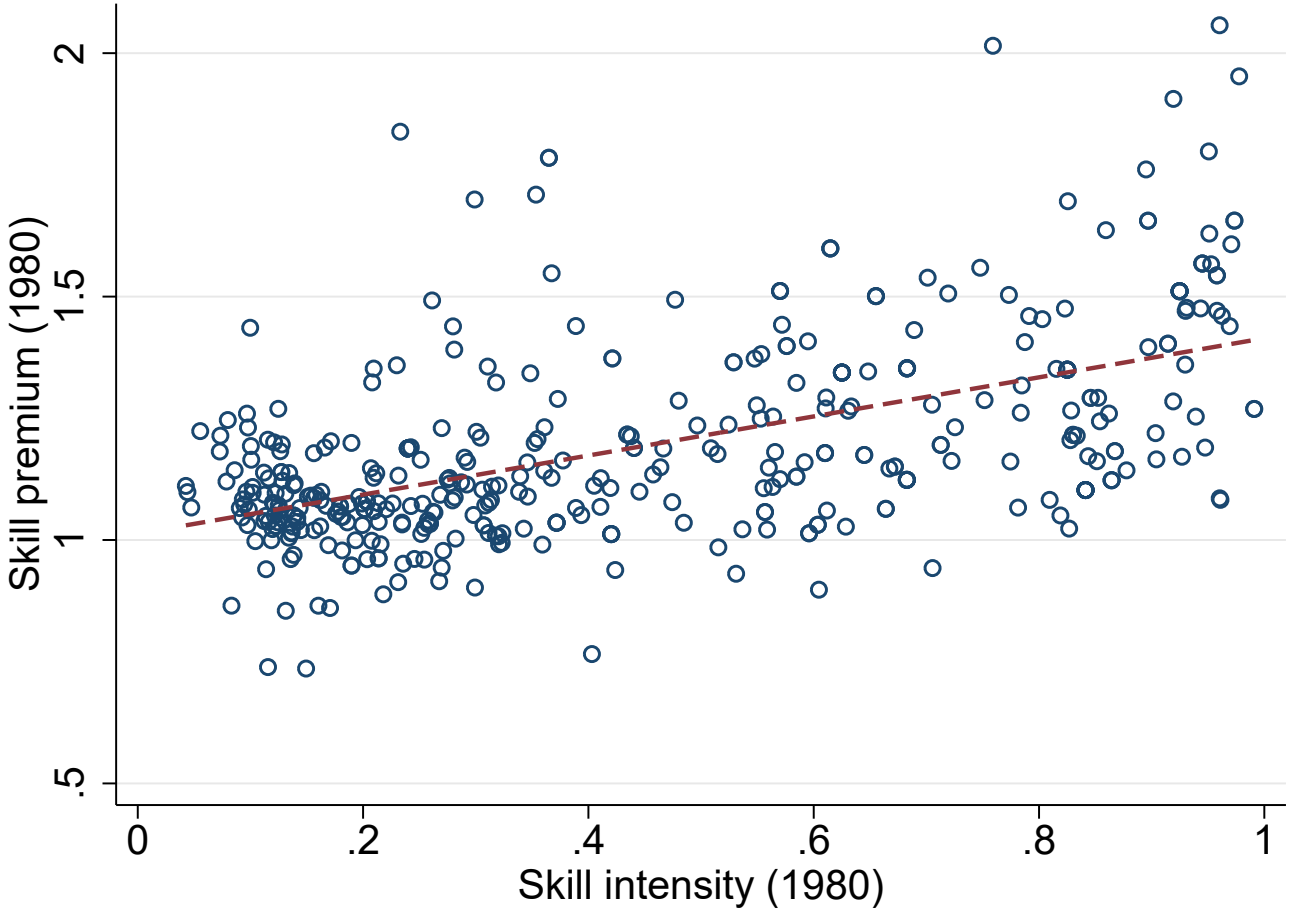
- The heterogeneous changes in x_{ot} generates 46% of actual increase in skill premium.
- Changes in λ_{ot} lead to an additional 32%p increase.
- Changes in κ_{ot} and ϵ_{ot} then reduce the increase in skill premium.
- Within-occupation margin accounts for most of the increase.
  - $\sigma > \rho$, so the relative demand *across* occupations responds more elastically than *within* an occupation.
  - $\eta > \rho$, so the elastic supply response reduces the wage and demand increases from a higher x_{ot}.

Scenario	Pre-1980	Post-1980	Patent Only	Demand	Within only	Across Only
Level	1.32	1.58	1.97	2.38	1.81	1.34
Relative	1	1.20	1.49	1.80	1.37	1.02

Table: Counterfactual Analysis -Results

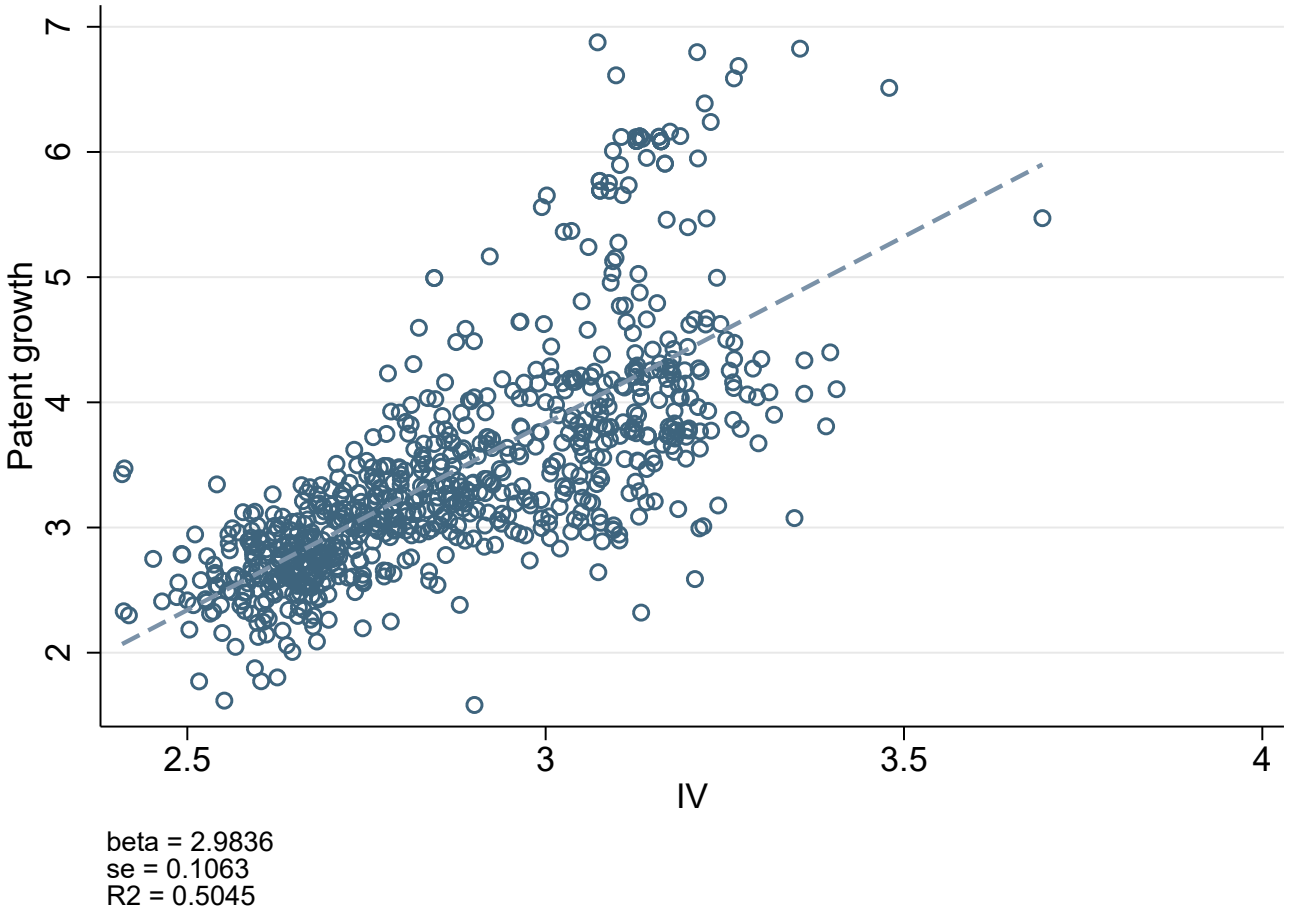
Appendix

Skill Premium and Skill Intensity



Skill premium = Average wage of skilled / Average wage of unskilled at occupation level. Skill intensity = share of skilled workers. Skilled = College graduate or higher back

Skill Premium and Skill Intensity



IV $= \sum_i s_{io} \sum_{j \neq i} s_{ij} \Delta \log P_{jt}$, where s_{ij} is patent class j's share in patent class i's references, and s_{io} is the share of patent class i in tools in occupation o. back

Example of tool innovation at occupation level

Occupation	Tools	Wikipedia found	Patents
Bakers	Automatic dough presses	Yes	15
Bakers	Bagel dividers	Yes	10
Bakers	Baking frames	Yes	10
Bakers	Baking sheets	Yes	5
Bakers	Balance scales	Yes	5
Bakers	Barquette molds	No	.

Then the tool innovation of bakers is $(15+10+10+5+5)/5 = 9$ back

Regression Result

$$\Delta y_{i,1990-2015} = \beta_0 + \beta_1 p_{i,1990-2015} + \epsilon_i$$

	(1) Wage	(2) Col. Pre.	(3) Emp.	(4) Wage	(5) Col. Pre	(6) Emp.
log(patent)	0.0182	0.0267	-0.126	0.00207	0.00445	0.0947***
	(0.0171)	(0.0158)	(0.0865)	(0.00234)	(0.00954)	(0.0174)
N	326	325	326	20467	14733	20616
sector fixed	no	no	no	yes	yes	yes

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Wage and college premium is weighted by wage bill in the initial period. Sector is NAICS 3 digit. Long difference between 2015 and 1990 back

Regression Result

$$\Delta y_{i,1990-2015} = \beta_0 + \beta_1 p_{i,1990-2015} + \epsilon_i$$

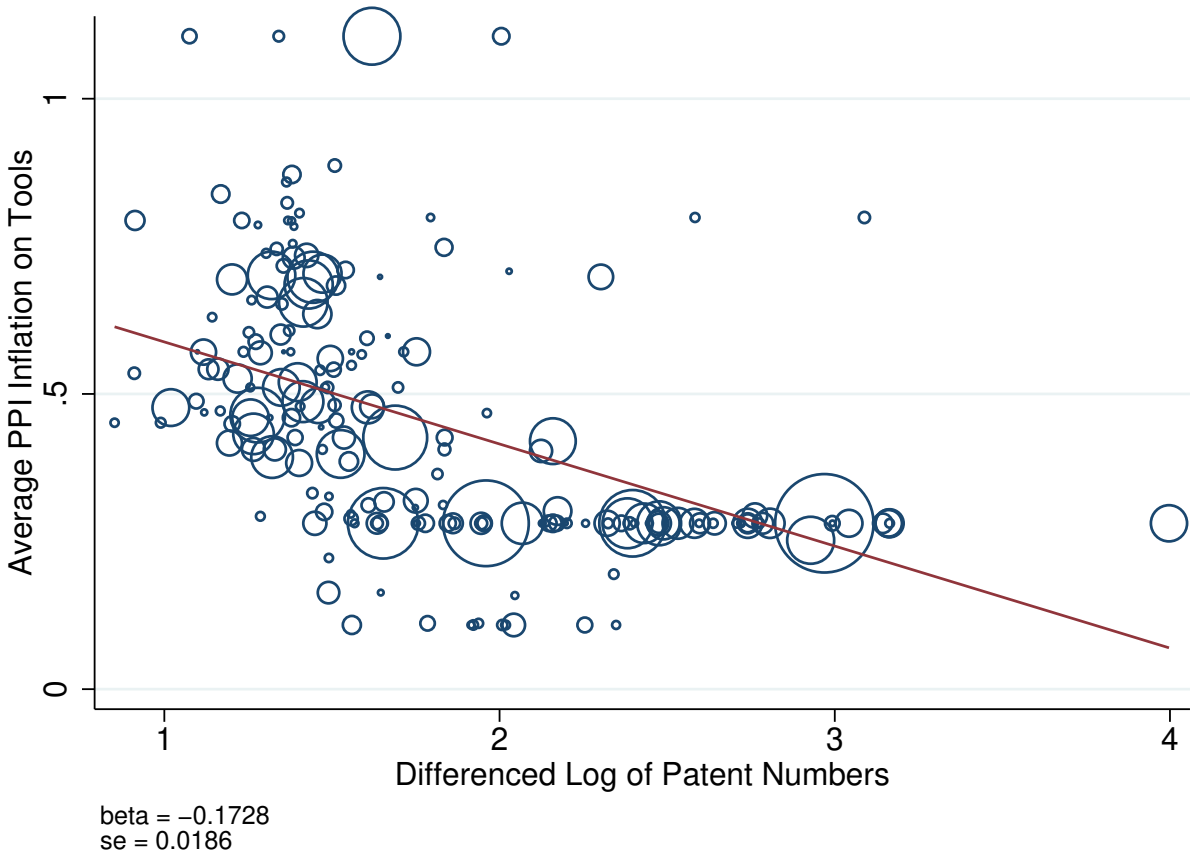
	(1) Wage	(2) Col. Pre.	(3) Emp.	(4) Wage	(5) Col. Pre	(6) Emp.
log(patent)	0.334^{***}	0.200^{***}	0.121	0.174^{***}	0.0711^{**}	0.448^{***}
	(0.0663)	(0.0502)	(0.156)	(0.00679)	(0.0244)	(0.0364)
N	326	325	326	20467	14733	20616
F (first stage)	50.76	50.57	50.76	3580.1	2621.7	3227.6
sector fixed	no	no	no	yes	yes	yes

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Wage and college premium is weighted by wage bill in the initial period. Sector is NAICS 3 digit back

More Innovation Reduces Price



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Effects of Cheaper Tools on Labor Market - OLS

$$\Delta y_{i,1990-2015} = \beta_0 + \beta_1 ppigr_{i,1990-2015} + \epsilon_i$$

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Wage	Col. Pre.	Emp.	Skill Int.	Wage	Col. Pre	Emp.	Skill Int.
PPI	-0.0621	-0.0702	-0.339	-0.111***	-0.194***	-0.0122	-0.252***	-0.00822*
	(0.126)	(0.0478)	(0.230)	(0.0205)	(0.0101)	(0.0334)	(0.0412)	(0.00406)
N	145	145	145	145	13288	9660	13387	13387
sector fixed	no	no	no	no	yes	yes	yes	yes

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Effects of Cheaper Tools on Labor Market - IV

$$\Delta y_{i,1990-2015} = \beta_0 + \beta_1 ppigr_{i,1990-2015} + \epsilon_i$$

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Wage	Col. Pre.	Emp.	Skill Int.	Wage	Col. Pre	Emp.	Skill Int.
PPI	-0.266	-0.361***	0.246	-0.181***	-0.533***	-0.313***	0.0680	-0.0208*
	(0.213)	(0.0900)	(0.395)	(0.0358)	(0.0214)	(0.0679)	(0.0836)	(0.00824)
N	145	145	145	145	13288	9660	13387	13387
F (first stage)	77.00	77.00	77.00	77.00	4208.6	3054.8	4241.0	4241.0
sector fixed	no	no	no	no	yes	yes	yes	yes

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

back

GMM Estimation - Details

$$\mathbb{E}\left[\left\{\Delta_t \log\left(\frac{\lambda_{ot}}{1-\lambda_{ot}}\right) - \Delta_t \mathbb{E}_t\left[\log\left(\frac{\lambda_{ot}}{1-\lambda_{ot}}\right)\right]\right\}\Delta_t \log z_{ot}^i\right]$$

$$= \mathbb{E}\left[\Delta_t \log\left(\frac{\lambda_{ot}}{1-\lambda_{ot}}\right) - \Delta_t \mathbb{E}_t\left[\log\left(\frac{\lambda_{ot}}{1-\lambda_{ot}}\right)\right]\right]\mathbb{E}\left[\Delta_t \log z_{ot}^i\right] = 0$$

$$\Rightarrow \frac{1}{N_o}\sum_o \left(\Delta_t \log\left(\frac{w_{ot}^s}{w_{ot}^u}\right) - \Delta_t \alpha_t - \frac{\sigma-1}{\sigma}(1-\delta)\Delta_t \log(x_{ot}) + \frac{1}{\sigma}\Delta_t \log\left(\frac{s_{ot}}{u_{ot}}\right)\right)\left(\Delta_t \log z_{ot}^i\right) \approx 0$$

$$\mathbb{E}\left[\left\{\log\left(\frac{\alpha_{ot}}{\alpha_{1t}}\right) - \Delta_t \mathbb{E}_t\left[\log\left(\frac{\alpha_{ot}}{\alpha_{1t}}\right)\right]\right\}\Delta_t \log z_{ot}^i\right]$$

$$= \mathbb{E}\left[\Delta_t \log\left(\frac{\alpha_{ot}}{\alpha_{1t}}\right) - \Delta_t \mathbb{E}_t\left[\log\left(\frac{\alpha_{ot}}{\alpha_{1t}}\right)\right]\right]\mathbb{E}\left[\Delta_t \log z_{ot}^i\right] = 0$$

$$\Rightarrow \frac{1}{N_o}\sum_o \left(\Delta_t \log\left(\frac{w_{ot}^s}{w_{1t}^s}\right) - \Delta_t \beta_t - \frac{\rho-\sigma}{\rho(\sigma-1)}\Delta_t \log\left(\frac{\theta_{ot}}{\theta_{1t}}\right) - \frac{\sigma-1}{\sigma}\Delta_t \log\left(\frac{x_{ot}}{x_{1t}}\right)\right.$$

$$\left. - \frac{\rho-\sigma}{\rho\sigma}\Delta_t \log\left(\frac{s_{ot}}{s_{1t}}\right)\right)\left(\Delta_t \log z_{ot}^i\right) \approx 0$$

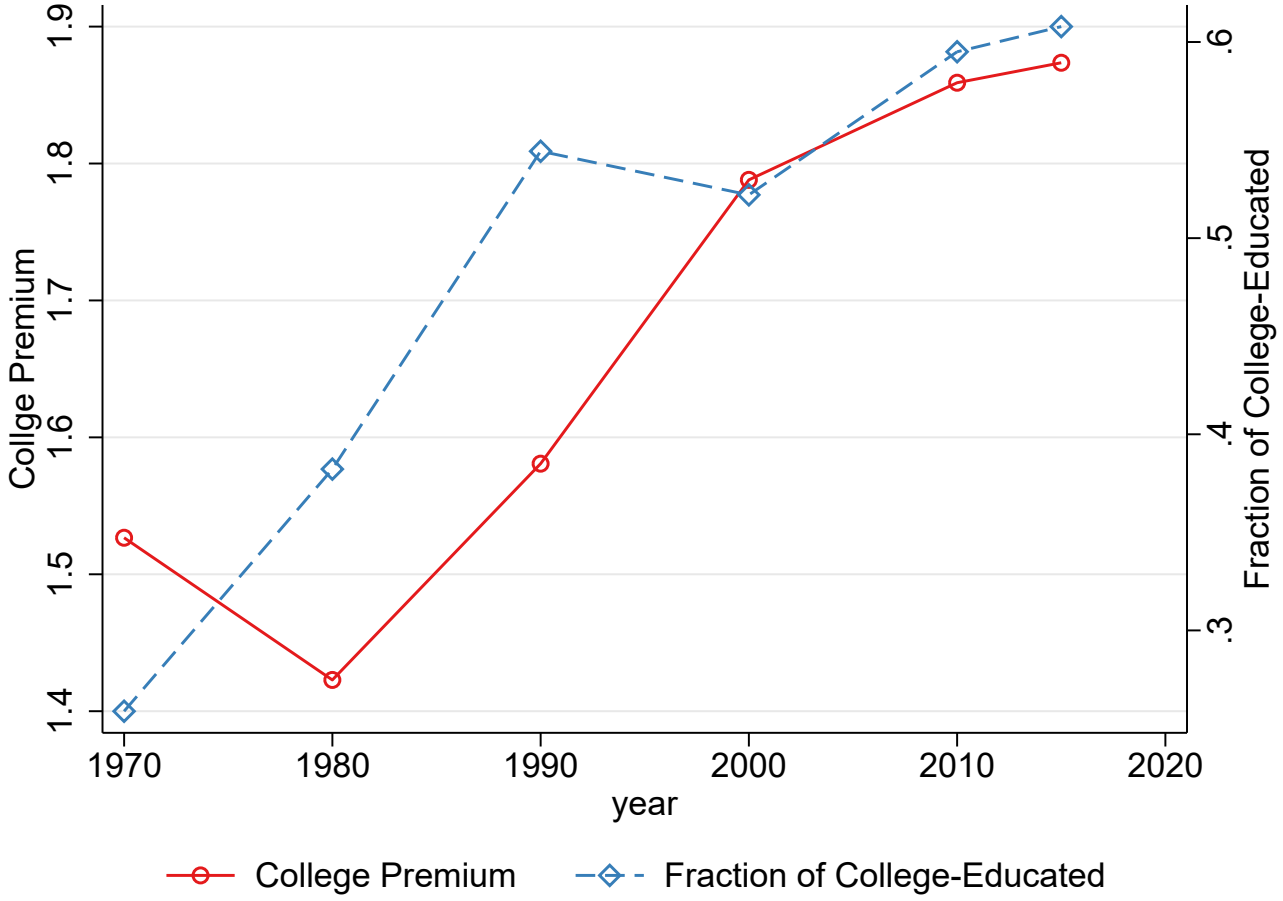
$$\mathbb{E}\left[\eta\left(\Delta_t \epsilon_{ot}^s - \Delta_t \epsilon_{1t}^s - \Delta_t \mathbb{E}_t\left[\epsilon_{ot}^s - \epsilon_{1t}^s\right]\right)\Delta_t x_{ot}\right] = \eta \mathbb{E}\left[\Delta_t \epsilon_{ot}^s - \Delta_t \epsilon_{1t}^s - \Delta_t \mathbb{E}_t\left[\epsilon_{ot}^s - \epsilon_{1t}^s\right]\right]$$

$$\times\, \mathbb{E}\left[\Delta_t \log x_{ot}\right] = 0$$

$$\Rightarrow \frac{1}{N_o}\sum_o \left(\Delta_t \log\left(\frac{s_{ot}}{s_{1t}}\right) - \Delta_t \gamma_t - \eta \Delta_t \log\left(\frac{w_{ot}^s}{w_{1t}^s}\right)\right)\left(\Delta_t \log x_{ot}\right) \approx 0$$

back

Rising Skill Premium and Supply of Skilled Workers



Skill Premium = Mean wage of college or above/Mean wage of high school or below

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Rising Skill Premium: Why?

- The literature has emphasized **skill-biased technical change**.
  - Define the aggregate production function as

$$Y = \left[(A_l L)^{(\sigma-1)/\sigma} + (A_h H)^{(\sigma-1)/\sigma} \right]^{\sigma/(\sigma-1)}$$

  - Assuming competitive wages, skill premium is simply

$$\omega = (A_h/A_l)^{(\sigma-1)/\sigma} (H/L)^{-1/\sigma}$$

Rising Skill Premium: Why?

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Table: Implied A_h/A_l, from Acemoglu (2002)

	Some college	College graduate
1950	0.146	0.219
1960	0.189	0.343
1970	0.199	0.485
1980	0.270	0.643
1990	0.357	1.064

Regression Result

$$\Delta y_{i,1990-2015} = \beta_0 + \beta_1 \Delta \log p_{i,1990-2015} + \epsilon_i$$

	(1) Wage	(2) Emp.	(3) Skill Pre.	(4) Skill Int.
patent	0.0845^{***}	0.144^{***}	0.0440^{***}	0.0165^{***}
	(0.0115)	(0.0411)	(0.00753)	(0.00329)
N	324	324	323	324

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Wage and employment is logged. Wage and college premium regression is weighted by wage bill in the initial period. Sector is NAICS 3 digit. other specification back

Construction of IV

$$\sum_{j \neq i} s_{ij} \Delta \log P_{jt} \quad \rightarrow \quad \Delta \log P_{it}$$

- s_{ij} is the citation share from technology class i to j
- We exclude technology sectors that are in the same NAICS 2 digit

IV Regression Result

$$\Delta y_{i,1990-2015} = \beta_0 + \beta_1 \Delta p_{i,1990-2015} + \epsilon_i$$

	(1) Wage	(2) Emp.	(3) Skill Pre.	(4) Skill Int.
patent	0.119^{***}	0.0600	0.0406^{***}	0.0297^{***}
	(0.0156)	(0.0551)	(0.0100)	(0.00449)
N	324	324	323	324
F (first stage)	410.8	410.8	409.6	410.8

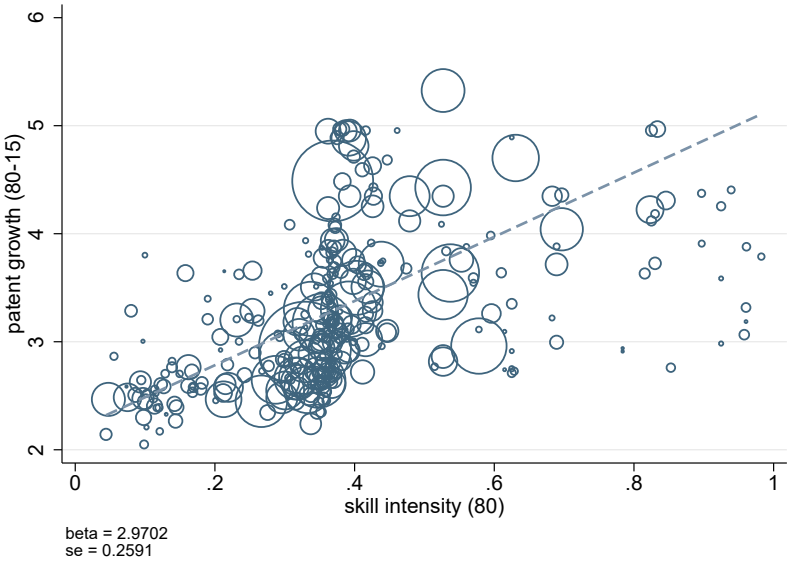
Standard errors in parentheses

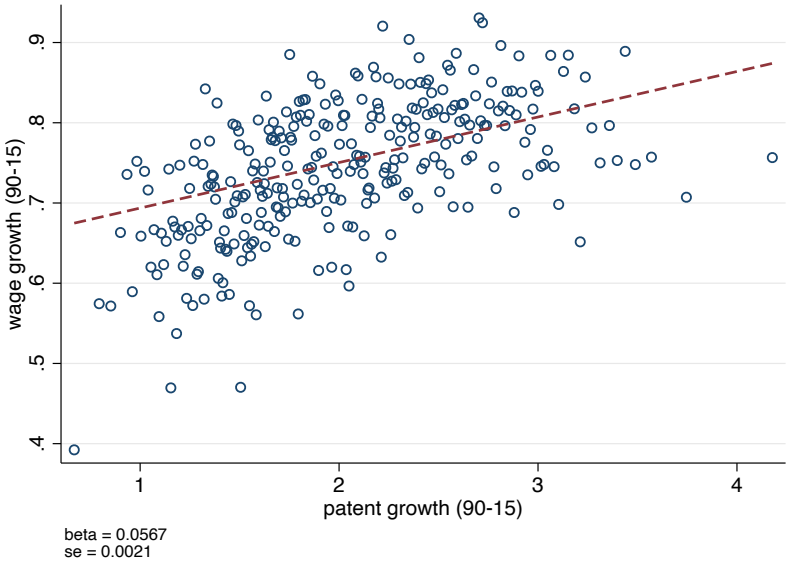
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Wage and employment is logged. Wage and college premium is weighted by wage bill in the initial period. Sector is NAICS 3 digit first stage other specification

★ How? Innovation reduces the price for tools. Innovation and Tool Price

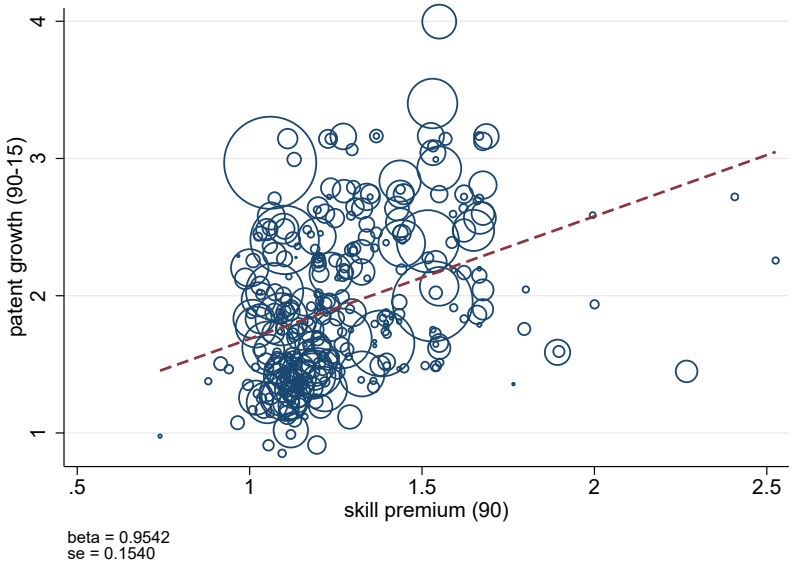
Skill-biased Innovation and Wage Increase

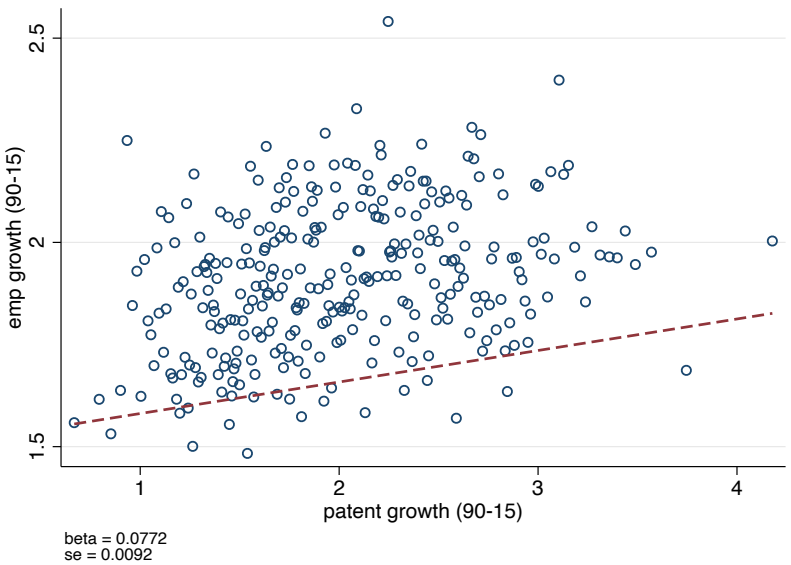




Skill intensity = skilled labor share at occupation level. Skilled = College graduate or higher. Wage growth is residual after controlling industry fixed effects. back

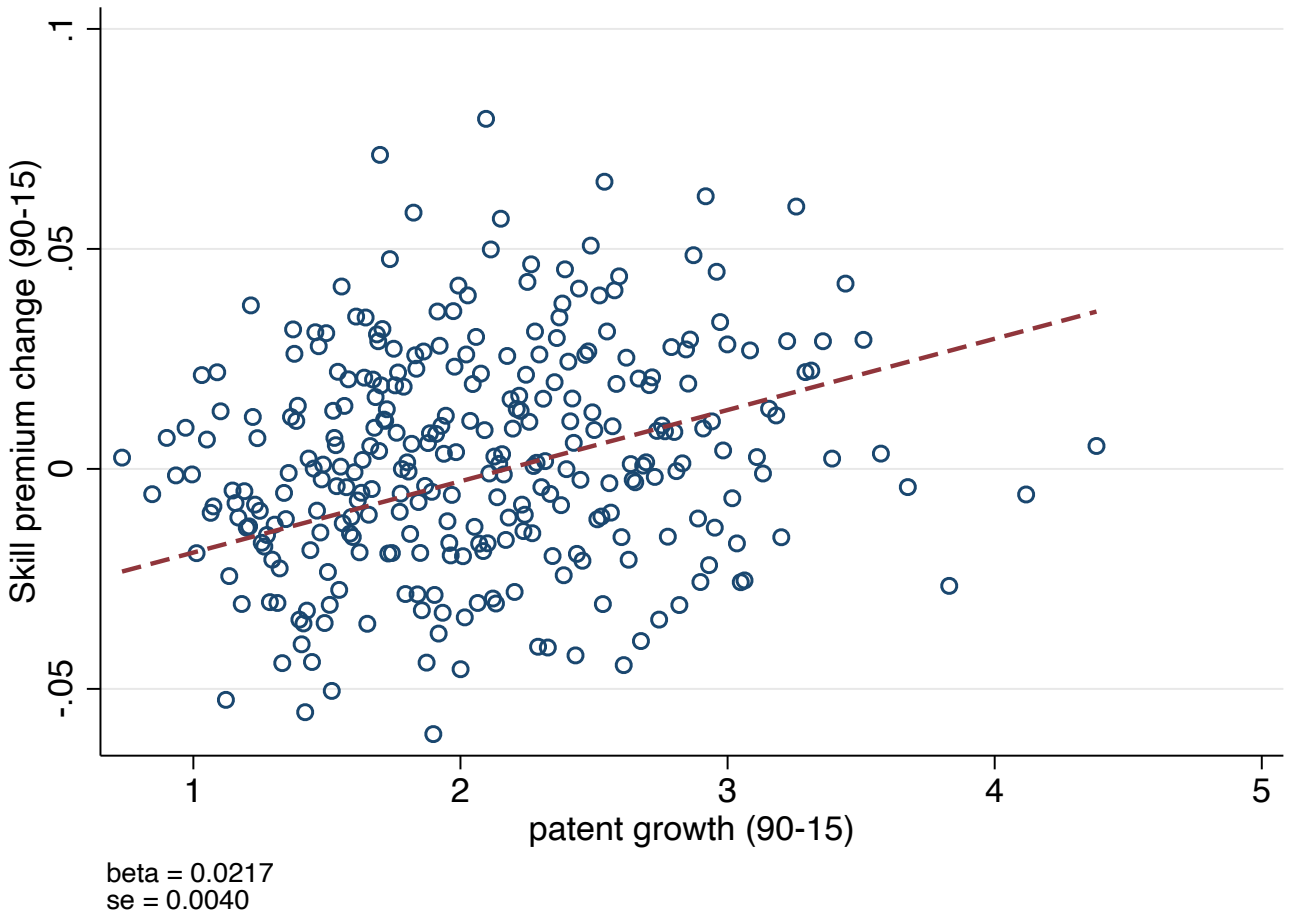
Skill-biased Innovation and Employment Increase





Skill premium = Average wage of skilled / Average wage of unskilled at occupation level. Skilled = College graduate or higher. Emp growth is residual after controlling industry fixed effects. back

Tool Innovation and Skill Premium Increase



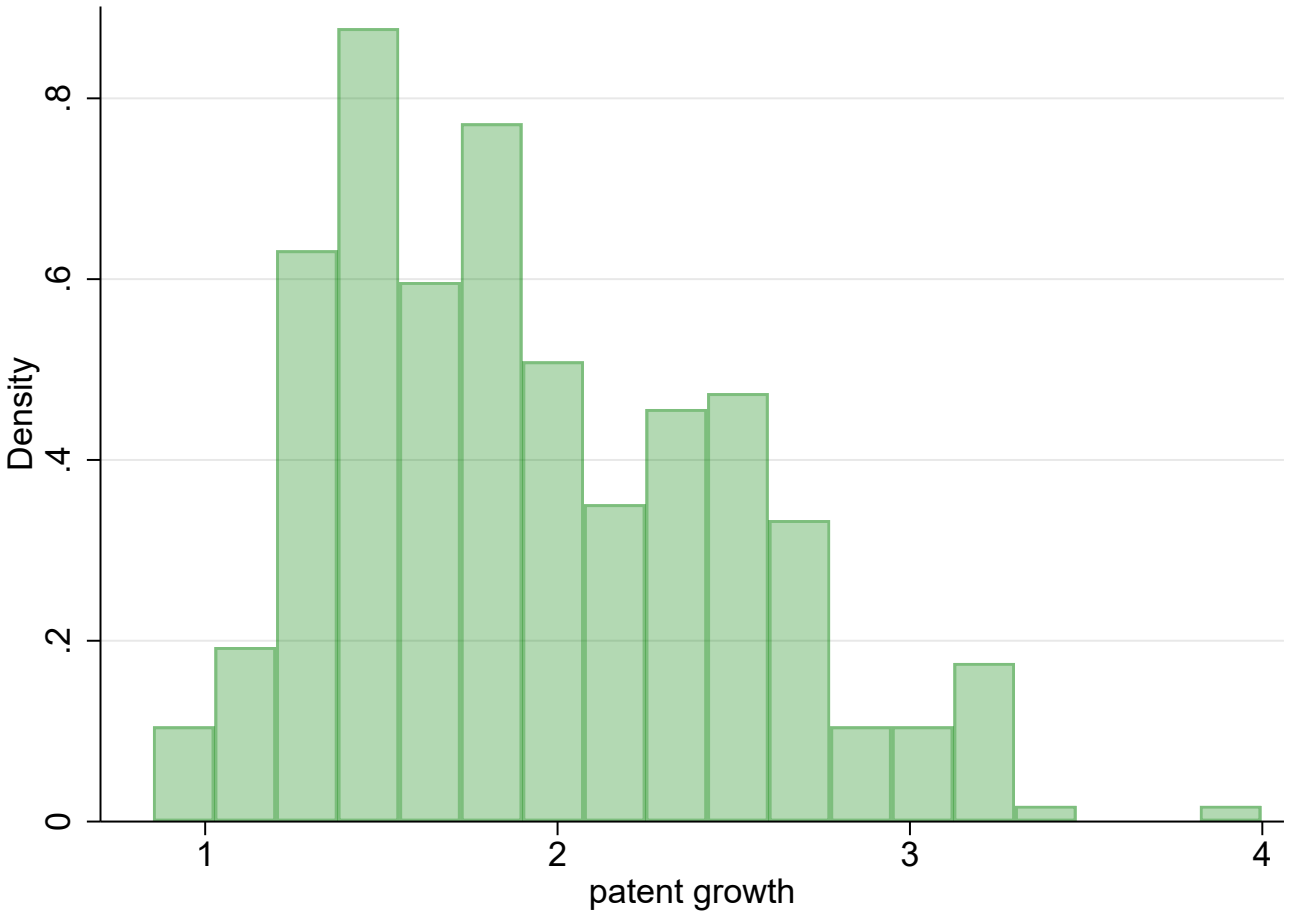
Skill premium = Average wage of skilled / Average wage of unskilled at occupation level.
Skilled = College graduate or higher. Residuals after controlling industry fixed effects. back

Compare with Webb's JMP

- This project
  - Measure innovation on **Tools** → Occupations
  - Study all tools provided by ONET
  - Assuming tools are **complements** to labor
- Webb's JMP
  - Measure technology on **Tasks** of occupations
  - Case study : AI, software, and robots
  - Assuming AI and robots are **substitutes** to labor at task level

back

Distribution of Patent growth at occupation level



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Regression Result

$$\Delta y_{i,1990-2015} = \beta_0 + \beta_1 \log p_{i,1990-2015} + \epsilon_i$$

	(1)	(2)	(3)	(4)	(5)	(6)
	Wage	Col. Pre.	Emp.	Wage	Col. Pre	Emp.
log(patent)	0.0182	0.0267	-0.126	0.00207	0.00445	0.0947***
	(0.0171)	(0.0158)	(0.0865)	(0.00234)	(0.00954)	(0.0174)
N	326	325	326	20467	14733	20616
sector fixed	no	no	no	yes	yes	yes

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Wage and employment is logged. Wage and college premium regression is weighted by wage bill in the initial period. Sector is NAICS 3 digit. back

Regression Result

$$\Delta y_{i,1990-2015} = \beta_0 + \beta_1 \log p_{i,1990-2015} + \beta_2 \log p_{i,1970-1990} + \epsilon_i$$

	(1)	(2)	(3)	(4)	(5)	(6)
	Wage	Col. Pre.	Emp.	Wage	Col. Pre	Emp.
log(patent)	0.0644***	0.0558***	-0.0426	0.0356***	0.0114	0.169***
	(0.0171)	(0.0165)	(0.0956)	(0.00253)	(0.0105)	(0.0194)
N	326	325	326	20467	14733	20616
sector fixed	no	no	no	yes	yes	yes

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Wage and employment is logged. Wage and college premium regression is weighted by wage bill in the initial period. Sector is NAICS 3 digit. back

Regression Result - using market value of patents

$$\Delta y_{i,1990-2015} = \beta_0 + \beta_1 \Delta \log p_{i,1990-2015} + \epsilon_i$$

	(1) Wage	(2) Emp.	(3) Skill Pre.	(4) Skill Int.
patent (value)	0.0458***	0.0898*	0.0316***	0.0112***
	(0.0103)	(0.0353)	(0.00650)	(0.00283)
N	324	324	323	324

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Wage and employment is logged. Wage and college premium regression is weighted by wage bill in the initial period. Sector is NAICS 3 digit. Market value of patent is from Kogan et al.(2017). back

IV Regression Result

$$\Delta y_{i,1990-2015} = \beta_0 + \beta_1 p_{i,1990-2015} + \epsilon_i$$

	(1) Wage	(2) Col. Pre.	(3) Emp.	(4) Wage	(5) Col. Pre	(6) Emp.
log(patent)	0.334^{***}	0.200^{***}	0.121	0.174^{***}	0.0711^{**}	0.448^{***}
	(0.0663)	(0.0502)	(0.156)	(0.00679)	(0.0244)	(0.0364)
N	326	325	326	20467	14733	20616
F (first stage)	50.76	50.57	50.76	3580.1	2621.7	3227.6
sector fixed	no	no	no	yes	yes	yes

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Wage and employment is logged. Wage and college premium is weighted by wage bill in the initial period. Sector is NAICS 3 digit. back

IV Regression Result

$$\Delta y_{i,1990-2015} = \beta_0 + \beta_1 \log p_{i,1990-2015} + \beta_2 \log p_{i,1970-1990} + \epsilon_i$$

	(1) Wage	(2) Col. Pre.	(3) Emp.	(4) Wage	(5) Col. Pre	(6) Emp.
log(patent)	0.114***	0.0708**	0.0570	0.0608***	0.0371**	0.202***
	(0.0228)	(0.0217)	(0.117)	(0.00335)	(0.0139)	(0.0234)
N	326	325	326	20467	14733	20616
F (first stage)	435.6	434.9	643.5	26698.8	19200.0	43609.6
sector fixed	no	no	no	yes	yes	yes

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Wage and employment is logged. Wage and college premium is weighted by wage bill in the initial period. Sector is NAICS 3 digit. back

IV Regression Result

$$\Delta y_{i,1990-2015} = \beta_0 + \beta_1 \Delta p_{i,1990-2015} + \epsilon_i$$

	(1) Wage	(2) Emp.	(3) Skill Pre.	(4) Skill Int.
patent (value)	0.136***	0.0981	0.0407***	0.0307***
	(0.0181)	(0.0556)	(0.0103)	(0.00479)
N	324	324	323	324
F (first stage)	214.0	214.0	213.3	214.0
sector fixed	no	no	no	no

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Wage and employment is logged. Wage and college premium is weighted by wage bill in the initial period. Sector is NAICS 3 digit. Market value of patent is from Kogan et al.(2017).

back